

SIGNUP CONVERSION ANALYSIS REPORT

Date: February 2025

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Objective: Investigate the 11% drop in signup conversion from Week 1 to Week 2

EXECUTIVE SUMMARY

The signup conversion rate decreased from 11.88% to 10.58% between Week 1 and Week 2, a 11% decline. Analysis reveals this drop was driven primarily by mobile users, whose conversion rate fell approximately 30% during this period. Desktop conversion remained stable, indicating a mobile-specific issue rather than a platform-wide problem.

Key Recommendation: Investigate mobile user experience changes, form functionality, and page load performance introduced in Week 2.

METHODOLOGY

Data Source: 10,000 user sessions across two weeks (5,000 per week)

Tools Used:

- MySQL for data storage and querying
- Python (pandas) for data processing
- SQL for conversion analysis
- Tableau for visualization

Metrics Analyzed:

- Overall conversion rate by week
- Conversion rate by device type
- Conversion rate by traffic source
- Week-over-week performance comparison

KEY FINDINGS

1. Overall Conversion Decline

- Week 1: 11.88% conversion rate
- Week 2: 10.58% conversion rate
- Change: -1.3 percentage points (-11% relative decline)

2. Mobile-Specific Drop (Root Cause)

- Mobile conversion dropped from ~11% to ~7.75% (-30%)
- Desktop conversion remained stable at ~14-15%
- Tablet conversion remained stable at ~5-6%
- Mobile accounts for ~60% of total traffic

3. Traffic Source Performance

- Organic traffic: Highest conversion (~13%)
- Direct traffic: Second highest (~11%)
- Social traffic: Moderate (~9%)
- Paid search: Lowest conversion (~8%)

4. Business Impact

At 5,000 weekly sessions:

- Expected signups (Week 1 rate): 594
- Actual signups (Week 2): 529
- Lost conversions: 65 signups per week
- Projected monthly impact: ~260 lost signups

ROOT CAUSE ANALYSIS

Primary Issue: Mobile conversion drop

Potential Causes to Investigate:

1. Mobile form UX changes (button size, field validation)
2. Page load time degradation on mobile
3. Mobile-specific code deployment in Week 2
4. Payment page rendering issues on mobile browsers
5. Increased friction in mobile checkout flow

Evidence Supporting Mobile-Specific Issue:

- Desktop performance unchanged
- Drop coincided with Week 2 start
- Consistent across all mobile traffic sources

RECOMMENDATIONS

Immediate Actions (Week 3)

1. **Review Week 2 deployments** - Identify any mobile-specific code changes
2. **Test mobile signup flow** - Complete end-to-end mobile signup on multiple devices
3. **Check mobile analytics** - Review bounce rates, form abandonment, error rates
4. **A/B test rollback** - If recent changes identified, test reverting them

Short-Term (2-4 weeks)

1. **Mobile UX audit** - Conduct usability testing on mobile signup flow
2. **Performance monitoring** - Implement mobile page load tracking
3. **Form analytics** - Add field-level tracking to identify drop-off points

Long-Term (1-3 months)

1. **Mobile-first redesign** - Optimize signup experience for mobile users
2. **Automated monitoring** - Set up alerts for conversion rate anomalies by device
3. **Regular testing** - Establish mobile device testing protocol for all releases

APPENDIX: SQL QUERIES USED

-- Overall Conversion by Week

```
SELECT
    week_number,
    COUNT(*) as total_sessions,
    SUM(signup_completed) as total_signups,
    ROUND(SUM(signup_completed) * 100.0 / COUNT(*), 2) as conversion_rate
FROM user_sessions
GROUP BY week_number;
```

-- Device Performance Analysis

```
SELECT
    device_type,
    week_number,
    COUNT(*) as sessions,
    SUM(signup_completed) as signups,
    ROUND(SUM(signup_completed) * 100.0 / COUNT(*), 2) as conversion_rate
FROM user_sessions
GROUP BY device_type, week_number
ORDER BY device_type, week_number;
```

****End of Report****